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EXPERIENCE

KTH Royal Institute of Technology

Research Assistant

May. 2026 – Present

- Deployed and optimized the **NVIDIA PersonaPlex-7B full-duplex speech model** for containerized GPU inference with CPU offloading.
- Integrated GPU-accelerated **Seed-VC** voice conversion into a containerized multi-service architecture.
- Migrated from a cloud-hosted **Kyutai Moshi** setup to a fully local infrastructure, resolving CUDA driver and PyTorch dependency conflicts.
- Developed a **WebSocket-based React frontend** implementing the PersonaPlex binary protocol with real-time **Opus audio encoding/decoding** and live transcription.
- Implemented **voice prompt conditioning** and real-time speech interaction features for multi-turn conversational AI workflows.
- Configured **HTTPS/WSS** communication with **self-signed certificates** to support secure browser-based microphone access.
- Designed asynchronous voice-conversion pipelines using **FastAPI** and **Starlette** BackgroundTasks with **HuggingFace** model caching and GPU resource allocation across containers.

Cefalo AS

Senior Software Engineer

Jan. 2025 – Aug. 2025

- Led **Generative AI** based projects using frameworks such as **Llamacpp** and **llamaindex**, and utilizing state-of-the-art large language models like **Whisper**, **OpenAI** and **Mistral** etc.
- Built **code modernization platform** to convert Mainframe applications into modern codebase, helping enterprises move away from legacy mainframes.
- Used **Agentic AI** to automate code generation and testing, enabling accurate transformation of legacy code while preserving core business logic.
- Designed **LangChain** to extract insights from legacy codebases and generate technical documentation, dependency graphs, and migration plans.
- Developed a **multi-tenant clinic portal** using **Blazor** for managing appointments, patients, and billing across organizations.
- Continuation of **.NET WPF** kiosk applications for patient check-in, appointment handling, and payments.
- Implemented automated UI testing using **FlaUI** for **.NET** desktop applications, improving regression coverage.
- Developed scalable **Python microservices**, **Dockerized sandbox for code execution**, and implemented distributed processing using **Celery** and **Redis**.
- Designed graph architecture for tracking code dependencies and used **MinIO** for storing transformation artifacts with proper version control.
- Created a real-time monitoring dashboard using **FastAPI** and **SSE**, also used **Grafana**, **Langfuse** to trace overall model and application flow.

Software Engineer

Jul. 2022 – Dec. 2024

- Designed and implemented **microservices architecture** using **Azure Functions** and **Service Bus** to enable event-driven communication between microservices. detailed specifications with clear project deliverables and timelines to meet strict deadlines.
- Worked on integrating **EHR** systems such as **Cambio Cosmic**, **DIPS**, **Semble**, and **Dentally**, connecting appointment and payment features.
- Developed **WPF** desktop applications for managing appointments and payments within EHR systems.
- Configured and managed **Redis Cache**, **Azure Blob Storage** to improve the performance and scalability of the software system
- Built backend services with **.NET Core** and **PostgreSQL** to support healthcare applications.
- Configured and managed **Azure API Gateway** to manage and secure API access and provide analytics on API usage, improving the security and usability of the software system.
- Implemented payment processing, handled **invoicing**, and managed **debt collection** workflows.

- Developed and maintained **CI/CD** pipelines using **Azure DevOps**, reducing deployment time and ensuring the availability and reliability of the software system.
- Architected high-performance data systems utilizing **Cosmos DB, MSSQL, and Entity Framework**
- Created and maintained technical documentation, user guides, and release notes, improving communication and knowledge sharing within the team and client.

Brainstation23 Ltd

Software Engineer

Jan. 2022 – Jun. 2022

- Worked closely with Project Managers and other developers to define parameters and plan detailed specifications with clear project deliverables and timelines to meet strict deadlines.
- Developed back-end Web services in **ASP.NET core** with **microservices** architectures.
- Developed reactive web application with **Rx.NET** and **Blazor**
- Worked with the latest cutting-edge technologies like **Docker, Kubernetes, GitHub-Actions** etc

Associate Software Engineer II

Nov. 2020 - Dec. 2021

- Designed client-side code with **Angular, Rxjs**, specifically known as **reactive web app**
- Collaborated in integrating diverse stack web UI applications featuring **web components** from React, Angular etc.
- Implemented real-time communication and data transmission through **SignalR** and **SSE**.

Dynamic Solvers Bangladesh Ltd

Dhaka, Bd

Junior Software Engineer

Nov. 2019 – Jun. 2020

- Developed high-performance API with **.Net Core**
- Developed Windows and Linux **worker services**
- Contributed to a map based mobile application in the Flutter
- Introduced and implemented a successful DevOps pipeline for the first time, resulting in a significant reduction in deployment failures, bugs, and overall deployment time.

PROJECTS

GpBot(personal project) | Python, PyTorch, Streamlit, Docker, BM25, BiLSTM Jan 2026 – Feb 2026

- Built a **bilingual FAQ chatbot** for Grameenphone that answers customer questions in English and Bangla(Romanized).
- Trained a **BiLSTM classifier** to categorize user queries into 14 intents to filter down knowledge base.
- Used **BM25 retrieval** to find the best answer from FAQs based on the predicted intent.
- Created a chat interface with **Streamlit** that handles conversation history.
- link: <https://gpbot.fahimabrar.com/>
- link: <https://github.com/syedfahimabrar/gpbot/>

Terminal | client: **ConveneGroup** | .NET Core, WPF, Blazor, Azure, FlaUI Apr 2024 – Sep 2025

- Built **multi-tenant healthcare platform** used by clinics for patient management, appointments, and billing workflows, with a shared backend supporting multiple organizations in isolation.
- Developed **.NET WPF kiosk** application for patient self-service, including check-in, appointment booking, and payment flows in clinical environments.
- Built a **Blazor**-based customer portal for clinic staff to manage patients, schedules, and administrative operations across multiple tenants.
- Extended the platform from Nordic region to UK market, adapting workflows and integrations to meet region-specific healthcare requirements.
- Integrated **external healthcare systems** such as Cambio Cosmic, DIPS, Semble, and Dentally for scheduling, billing, and patient data synchronization.
- link: <https://convenegroup.com/solutions/terminals/indoor-terminal/>

Meetsum | client: **Cefalo** | Python, NextJS, Runpod, CosmosDB, Llamacpp, Whisperx, Mistral Jan 2024 – Present

- **Led GenAI** project development, overseeing **task management**, ensuring timely delivery, and coordinating efforts across the development team.
- Developed advanced **transcription** and **summarization** features using fine-tuned large language models such as OpenAI's **Whisper** and **Mistral-7B**.

- Implemented **full-stack** development, developing audio recording features with **RecordRTC** and utilizing **Next.js** for front-end development to create a seamless user interface that works with any meeting platform.
- Integrated **Azure Blob Storage** for efficient handling and storage of large audio datasets.
- Used **Llamacpp** to serve and query **LLM** efficiently, successfully serving **quantized** models to increase GPU memory efficiency while maintaining relevant results.
- Optimized **scalability** and **performance** through **serverless architecture** using **Runpod** and **Docker**.

Jengax | client: Spenn AS | .NET Core, Azure Function, CosmosDB, Redis, Azure Service Bus Jul 2022 – Dec 2023

- Contributed to develop subscription-based microservice API with **Azure APIM**, **Azure Function**, **Azure Service Bus** and **Azure Active Directory**
- Contributed to apply rate limit in the current system synchronized with third party systems
- Developed token bucket using semaphore to avoid duplicate auth request to external system
- Developed **CI/CD** pipeline with **Azure Devops** and **Powershell** to automate the deployment in dev,uat,test and prod environment
- Contributed to develop **Iaac** for the project with help of **Bicep**
- Developed and designed efficient database containers, aligned with intricate business logic, to optimize the **SAAS** application's performance and scalability through **sharding**.
- Developed notification services to notify important messages through **webhooks** and socket URL with **SignalR**
- link: <https://www.jengax.com/>

Informatiq(Informatiq AS) | .Net Core, Angular, RxJS, Docker, Azure Devops

Nov 2020 – Jun 2022

- Contributed to development of microservice-based api using **.NET Core**
- Designed and developed reactive front-end application with **Angular**, **RxJS** and **Blazor**
- Implemented continuous delivery using **Azure Devops** to deploy the system upon new release
- Collaborated closely with clients to efficiently manage and resolve new tasks in Azure DevOps **Kanban** board, ensuring seamless workflow and timely resolution of issues.
- link: <https://informatiq.no/>

TECHNICAL SKILLS

Languages: C#, TypeScript, Javascript, MSSQL, Powershell, Python, R, Bicep

Frameworks: Asp.net Core, .Net Core, FastAPI, Angular, Blazor, RxJS, Entity Framework, Microsoft Graph, SignalR, XUnit

DevOps/IaC Tools: Git, Docker, TravisCI, Github Actions, Azure Devops, Bicep, Pulumi

Design Patterns: SOLID, Singleton, Factory, Mediatr, CQRS

Database: MSSQL, PostgreSQL, CosmosDB, Redis, Pinecone, Neo4j

AI: Pytorch, Scikit-learn, Numpy, pandas, matplotlib, dplyr, ggplot2, tidyverse, OpenAI, vLLM, Langchain, LlamaIndex, Autogen, CrewAI, Llamacpp, LlamaIndex, PEFT, RAG, MCP, Pandas

Observability : Azure App Insights, Grafana, Langfuse, WnB

Auth: Auth0, Azure AD (Microsoft EntraID), OAuth2, OpenID Connect, JWT, RBAC

Cloud: Azure, AWS, Kubernetes

EDUCATION

Stockholm University

Master's of Science in AI & Language

Stockholm, SE

Sep. 2025 – Present

Daffodil International University

Bachelor of Science in Computer Science & Engineering

Dhaka, BD

Jan. 2016 – Jan 2020

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals

available at <https://www.credly.com/badges/540a0fb3-a730-484d-92d6-702ecf55f9cc>

Dhaka, BD

Jan. 15,2022

Generative AI with Large Language Models by DeepLearning.AI

available at <https://www.coursera.org/account/accomplishments/verify/7F6YGZBG8AY5>

Dhaka, BD

Feb. 28,2024